An Introduction to the cov vs pp problem

Lecture 2: Pseudopower calculus and applications

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The pseudopower calculus

1: No Holes

Let $\aleph_0 \leq \sigma < \theta$ be regular, and put $\Gamma = \Gamma(\theta, \sigma)$.

Assume $\sigma \leq \mathsf{cf}(\lambda) < \theta < \lambda$.

Then $\mathsf{PP}_\Gamma(\lambda)$ is an initial segment of the regular cardinals above λ.

In particular,

$$\lambda < \kappa < \mathsf{pp}_\Gamma(\lambda), \qquad \kappa \in \mathrm{Reg} \quad \Longrightarrow \quad \kappa \in \mathsf{PP}_\Gamma(\lambda).$$

Attainment of the supremum $\mathsf{pp}_\Gamma(\lambda)$ is a separate issue.

2: Inverse Monotonicity

Let $\aleph_0 \leq \sigma < \theta$ be regular, and put $\Gamma = \Gamma(\theta, \sigma)$.

Suppose

$$\theta < \lambda < \mu \leq \mathrm{pp}_\Gamma(\lambda), \qquad \mathrm{cf}(\lambda), \mathrm{cf}(\mu) \in [\sigma, \theta).$$

Then

$$\mathrm{PP}_\Gamma(\mu) \subseteq \mathrm{PP}_\Gamma(\lambda),$$

and hence

$$\mathrm{pp}_\Gamma(\mu) \leq \mathrm{pp}_\Gamma(\lambda).$$

Let $\aleph_0 \leq \sigma < \theta$ be regular, and put $\Gamma = \Gamma(\theta, \sigma)$.

Suppose $\sigma \leq \mathsf{cf}(\lambda) < \theta < \lambda$, and let $\kappa > \lambda$ be regular.

If, for arbitrarily large singular $\mu < \lambda$,

$$\sigma \leq \mathsf{cf}(\mu) < \theta \qquad \text{and} \qquad \kappa \in \mathrm{PP}_\Gamma(\mu),$$

then

$$\kappa \in \mathrm{PP}_\Gamma(\lambda).$$

4: Compression

Let $\aleph_0 \leq \sigma < \theta$ be regular, and suppose $\sigma \leq \mathrm{cf}(\lambda) < \theta < \lambda$.

Assume that for every sufficiently large singular $\mu < \lambda$,

$$\sigma \leq \mathrm{cf}(\mu) < \theta \quad \Longrightarrow \quad \mathrm{pp}_{\Gamma(\theta,\sigma)}(\mu) < \lambda.$$

Then

$$\mathrm{PP}_{\Gamma(\theta,\sigma)}(\lambda) = \mathrm{PP}_{\Gamma(\mathrm{cf}(\lambda)^+,\sigma)}(\lambda),$$
$$\mathrm{pp}_{\Gamma(\theta,\sigma)}(\lambda) = \mathrm{pp}_{\Gamma(\mathrm{cf}(\lambda)^+,\sigma)}(\lambda).$$

Thus representations can be compressed sets of cardinality $\mathrm{cf}(\lambda)$ while preserving σ-completeness.

In particular, the hypothesis holds when λ is strong limit.

An application

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Put $P = \mathsf{pp}(\lambda)$ and suppose $\mathsf{cf}(P) \leq \lambda$.

Then P is singular and $\mathsf{cf}(P) < \lambda$. Choose a regular cardinal θ such that

$$\max\{\mathsf{cf}(\lambda), \mathsf{cf}(P)\} < \theta < \lambda,$$

and put $\Gamma = \Gamma(\theta, \aleph_0)$.

The proof

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But P^+ is representable at P using using a set of size $\mathrm{cf}(P) < \theta$. Thus

$$P^+ \leq \mathrm{pp}_\Gamma(P) \leq P,$$

a contradiction.

An application we will need later

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If λ is strong limit singular and $P = \mathrm{pp}(\lambda)$, then

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$$\mathrm{pp}_\Gamma(\lambda) = \mathrm{pp}(\lambda) = P.$$

The proof

For $\lambda \leq \mu \leq P$ with $\mathsf{cf}(\mu) = \aleph_1$, compression and inverse monotonicity give

$$\mathsf{pp}_{\Gamma(\aleph_1)}(\mu) \leq \mathsf{pp}_\Gamma(\mu) \leq P.$$

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The general cov vs. pp theorem gives

$$\begin{aligned} &P + \mathrm{cov}(P, \lambda, \aleph_2, \aleph_1) \\ &\quad = P + \sup_{\substack{\lambda \leq \mu \leq P \\ \mathsf{cf}(\mu) = \aleph_1}} \mathrm{pp}_{\Gamma(\aleph_1)}(\mu) = P. \end{aligned}$$

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Since $\lambda < P$, the covering number is at least P. Thus

$$\mathrm{cov}(P, \lambda, \aleph_2, \aleph_1) = P.$$

Cov Vs pp Problem

Four cardinals

Fix a strong limit singular cardinal λ with $\mathrm{cf}(\lambda) = \aleph_0$.

We will compare four cardinals $\kappa_1, \ldots, \kappa_4$. Two are already familiar:

$$\kappa_1 = \mathrm{pp}(\lambda),$$

$$\kappa_3 = \mathrm{cov}(\lambda, \lambda, \aleph_1, 2) = \lambda^{\aleph_0}.$$

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Our original question is whether

$$\kappa_1 = \kappa_3.$$

The second cardinal

Put $R = \lambda \cap \mathrm{Reg}$.

Definition

κ_2 is the least size of a family $F \subseteq \prod R$ such that whenever $A \subseteq R$ is countable and $g \in \prod A$, there are $f \in F$ and $\mu < \lambda$ such that

$$g(a) < f(a) \qquad \text{for all } a \in A \setminus \mu.$$

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$\kappa_1 \leq \kappa_2 \leq \kappa_3$ is easy.

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A family $\mathcal{A} \subseteq [\kappa_2]^{\aleph_1}$ is *strongly almost disjoint* if distinct $A, B \in \mathcal{A}$ satisfy

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Maximality means that for every $X \in [\kappa_2]^{\aleph_1}$ there is $A \in \mathcal{A}$ such that

$$|X \cap A| \geq \aleph_0.$$

Comparing κ_2 and κ_4

Maximality gives $\kappa_2 \leq \kappa_4$.

Equivalently, κ_4 is the least size of a maximal family $\mathcal{F} \subseteq {}^{\omega_1}\kappa_2$ whose distinct members satisfy

$$\left|\{i < \omega_1 : f(i) = g(i)\}\right| < \aleph_0.$$

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Theorem (Shelah)

If $\kappa_2 < \kappa_4$, there are finite sets $\mathfrak{a}_i \subseteq \mathrm{Reg} \cap \kappa_2$ for $i < \omega_1$ such that

$$\max \mathrm{pcf}\left(\bigcup_{i \in u} \mathfrak{a}_i\right) > \kappa_2$$

for every infinite $u \subseteq \omega_1$.

Whether such a configuration is consistent is a major unsolved question in pcf theory.

Proving $\kappa_3 \leq \kappa_4$

Fix $\bar{f} = \langle f_\alpha : \alpha < \kappa_2 \rangle$ witnessing the definition of κ_2.

Let $\mathcal{A} \subseteq [\kappa_2]^{\aleph_1}$ be maximal strongly almost disjoint, with $|\mathcal{A}| = \kappa_4$.

Choose an increasing sequence of cardinals $\bar{\mu} = \langle \mu_n : n < \omega \rangle$ with

$$\aleph_2 \leq \mu_0, \qquad \sup_{n<\omega} \mu_n = \lambda.$$

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All models and hulls are taken in a sufficiently large $H(\chi)$ with fixed Skolem functions and $\lambda, \bar{f}, \bar{\mu}$ named as constants.

The covering family

For $A \in \mathcal{A}$ and $n < \omega$, put

$$H_{A,n} = \mathbf{Hull}\big((\mu_n + 1) \cup A\big).$$

Then $|H_{A,n}| \leq \mu_n < \lambda$.

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$$\mathcal{C} = \{H_{A,n} \cap \lambda : A \in \mathcal{A},\ n < \omega\}.$$

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We have $|\mathcal{C}| \leq \kappa_4$.

Fix a countable $Y \subseteq \lambda$. We must find a member of $\mathcal{C}$ containing Y.

Building the tower

Build a continuous increasing tower $\langle N_i : i < \omega_1 \rangle$ of countable elementary submodels, with $Y \subseteq N_0$.

At stage i, use κ_2 to choose $\alpha_i < \kappa_2$ and $n(i) < \omega$ with

$$\sup(N_i \cap \rho) < f_{\alpha_i}(\rho)$$

for $\rho \in N_i \cap R$ above $\mu_{n(i)}$. Put α_i into N_{i+1}.

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No function in N_i can satisfy this inequality. Thus $\alpha_i \notin N_i$, and the α_i's are distinct.

Using maximality

Choose an uncountable $I \subseteq \omega_1$ and $n < \omega$ such that

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Choose $i_0 < i_1 < \cdots$ from I with $\alpha_{i_m} \in A$, and put

$$\delta = \sup_{m<\omega} i_m < \omega_1.$$

Comparing the models

Put $N = N_\delta$ and $M = H_{A,n} \cap N$. Then $M \prec N$, and the models agree below $\mu_n + 1$.

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By our earlier argument,

$$Y \subseteq N \cap \lambda = M \cap \lambda \subseteq H_{A,n} \cap \lambda \in \mathcal{C}.$$

Thus $\kappa_3 \leq \kappa_4$.

Summary

If λ is a singular strong limit cardinal of countable cofinality then $\kappa_1 \leq \kappa_2 \leq \kappa_3 \leq \kappa_4$